

Data Engineering Case Study: Data Engineering and Master Data Management MVP for Retail Orders Analytics

Scenario:

Your client is a retail company which sells a variety of products online. Their finance department want to build dashboards for descriptive analysis of company's orders and sales in terms of time periods, products, customers, and regions. Business wants to have clean high-quality data on monthly basis, extracted into Data Warehouse from the source system after all corrections. They also want to have cleansed master data for product, customer, and geographical data domains. Business follows agile methodology and prefers to start this project with MVP to understand as soon as possible pitfalls associated with their data.

Objective:

Deloitte is being involved at data engineering and data quality streams of the project. As final state Deloitte need to deliver high quality data in Data Warehouse which can be used for descriptive analysis of Sales Orders.

You as a Data Engineer of Deloitte team will need to design and implement MVP solution using open-source software and libraries following project plan described in the following paragraphs. As part of it you will design high-level technical architecture, analyze data inconsistencies, define golden records for key master data elements, build a robust ETL data pipeline that takes raw transactional data and transforms it into a structured format suitable for analytics. (You will be able to find hints for key steps below)

Data Source Description:

Client's source system pushes data into SFTP server.

Full copy of the data from SFTP server shared with you into zip archive "Case_Study_202309_Data.zip"

Key agreements with the source system:

- File Type: CSV with ANSI encoding
- Each csv has Orders for 1 month
- Data for specific month is pushed till the end of next months. (E.g., Orders for Aug'22 will be available in container till EOM Sep'22)
- Data for specific month can be pushed multiple times during next month as there can be corrections in Orders. Each time source system extracts and pushes all data for a specific month! (Refer to example below)
- Only successfully delivered orders are pushed (without returns)

File names mask on SFTP: YYYYMM_Orders_YYYY_MM_DD_HH_MM_SS

Month of orders	TimeStamp of file load
 202210_Orders_2022_11_01_00_08_55.csv	 2022_11_01_00_08_55.csv
 202211_Orders_2022_12_02_21_54_39.csv	 2022_12_02_21_54_39.csv
 202211_Orders_2022_12_05_21_17_28.csv	 2022_12_05_21_17_28.csv
 202211_Orders_2022_12_08_15_01_57.csv	 2022_12_08_15_01_57.csv
 202211_Orders_2022_12_11_08_20_07.csv	 2022_12_11_08_20_07.csv

Example: `202211_Orders_2022_12_02_21_54_39` - file contains all orders from the source system for Nov'22 and it was loaded from source system on 2nd Dec'22 at 21:54:39.

Files Structure:

Column	Description
Row ID	Row number in files
Order ID	Unique Order Id
Order Date	Order creation date by customer
Ship Date	Order shipment date
Ship Mode	Order priority type (Same day, First Class, ...)
Customer ID	Unique Customer Id
Customer Name	Customer Name
Segment	Consumer Segment (Consumer, Corporate, ...)
Country	Country of delivery
City	City of delivery
State	State of delivery
Postal Code	Postal Code of delivery
Region	Region of delivery
Product ID	Unique Id of ordered product
Category	Product category
Sub-Category	Product sub-category
Product Name	Product name
Sales	Sales amount in USD
Quantity	Qty of product in order
Discount	Discount
Profit	Profit from delivery of the product

Your key tasks and deliverables:

! Follow instructions for each deliverable file names and types!

! Share all deliverables in 1 archive!

You are free to suggest and use any open-source tools, frameworks & program languages

Phase 1 - Define & Design

At first phase you need to analyze requirements and data sources, do high level design of your solution, and make data model for data warehouse.

1. Reporting Requirement Analysis:

Business sent their requirements what should be presented on MVP dashboard:

- a) Dynamics of avg price for best-selling products
- b) Sales dynamics by states
- c) Best customers in segments
- d) Top/Bottom products according to sales by state
- e) Top/Bottom products according to avg delivery time by state.

Task 1: (10 min)

- Define glossary for KPIs which are needed for dashboard according to mentioned requirements.

E.g.:

KPI	Description	Formula	Granularity
Max Profit	Maximum product profit for month	MAX(Profit) by Month(Order Date)	Month, Product

Deliverable 1:

- File Name: Task_1_KPI_Glossary.csv
- File Type: CSV
- File Headers: KPI, Descriptions, Formula, Granularity

2. Solution High Level Design:

Task 2: (30 min)

- Design technical solution for MVP
- You need to describe how will you ingest data, process data, store data, build ETL pipelines and automate execution
 - a) Analyze Data Source & Describe how will you ingest & process files on regular basis
 - b) Propose Technical Architecture for MVP (software & tools)
 - c) Describe structure of target DWH (layers or schemas)
 - d) Describe how will you automate data pipelines
 - e) Describe how will you analyze and deal with inconsistencies in data

Deliverable 2:

- File Name: Task_2_HLD.pptx

- File Type: pptx
- Key Slides to consider: Technical Architecture, Data Processing Flow, Structure of DWH, Inconsistency Analysis & Cleansing (You may add any relevant slides at your discretion)

(Hint: You can consider ETL/ELT/Orchestrator tool and Data Warehouse for end-to-end data flow)

3. Data Modelling:

Do logical data modelling of your data warehouse target tables which will be consumed by Users and Analytical Applications (e.g., business intelligence tools).

Considerations:

- a) Do it according to your understanding of data required for the dashboard
- b) Keep in mind that business wants to have cleansed master data separated from transactional data
- c) Reduce redundancy in stored data and do normalization according to third normal form (3NF)

Task 3: (20 min)

- Make entity relationship diagram (ERD) of your model

(Hint: You can use free modelling tools which might help you with further steps, or just do it in Power Point)

Deliverable 3:

- File Name: Task_3_ERD.jpg
- File Type: Image (You can use any popular formats)

4. DWH Physical Design:

Do physical design of your data warehouse with SQL statements

- a) Create all necessary databases and schemas in DWH (according to High level design document)
- b) Make a dimensional model based on suggested Logical Data Model on 3rd step
- c) Create all necessary table structures with suitable data types in DWH (according to your dimensional model and analysis of the files from the source system)
(Hint: You might consider doing high-level data profiling of the source system files during current step for better typification)
- d) Add key necessary constraints to your tables aiming on data interoperability

Task 4: (20 min)

- Create one text file with all DDLs scripts for your data warehouse

Deliverable 4:

- File Name: Task_4_DDL.txt
- File Type: txt
- File Structure: DDL scripts for your tables in Data Warehouse

Phase 2 - Process and Deliver

During that phase you need to process data and store it in data warehouse according to your design.

5. ETL and Data Cleansing:

Build ETL pipeline for extraction, processing, and cleaning of data from data source and load data to DWH data marts.

You can consider following steps:

- a) Data Extraction (data connection and extraction)
- b) Data Profiling (data examination and typification)
- c) Data Cleansing (data cleansing in terms of inconsistencies)
- d) Data Processing and Normalization (according to designed HLD and Data Model)
- e) Master data qualifying (defining golden records for key master data elements)
- f) Data Loading in Data Marts
- g) Automation & Triggers (creating robust pipeline for data processing and adding triggers for regular execution)

(Hint: some inconsistencies can be solved only in cooperation with business or SMEs, so you can skip rows in your data processing pipeline with those inconsistencies and generate report which can be shown to SMEs for decision making)

Task 5: (120 min)

- Create a report with all data inconsistencies which you found during data processing. You need to mention number of rows affected by each type of inconsistency which you found and examples of those rows in raw data.
 - For those inconsistencies which can be handled technically without additional clarifications, mention how you handled them
 - For those inconsistencies which need additional clarifications – mention how you suggest solving them
 - Extract examples of rows from raw files which are affected by each type of inconsistency; Highlight exact values with issues.
- Design and generate data quality report for business and SMEs with all inconsistencies which need additional inputs for resolution. We suggest loading in that report all affected

rows which cannot be processed and update it after each pipeline execution (So, business will have understanding what data and why is not presented in data marts and reports)

(Hint: Example of inconsistency - different data types in one column)

(Hint: There are 5 different types of inconsistencies in shared data)

Deliverable 5:

- File Name: Task_5_Inconcistencies_Analysis.xlsx
- File Type: xlsx
- File Structure: Excel file with 3 Sheets:
 - 1) Summary of inconsistency analysis with calculation of number of affected rows and your comments how you suggest handling it
 - Sheet Name: Inconsistencies_Summary
 - Columns: Inconsistency Type - Description - Suggestion to handle - Distinct Count of [Row Id]
 - 2) Examples of rows from source files with inconsistencies with issues highlighting
 - Sheet Name: Inconsistencies_Examples
 - Columns: Inconsistency Type - <Columns from source files>
(It's enough to have 2 rows for each type of inconsistency)
 - 3) Data Quality Report for business and SMEs
 - Sheet Name: Quality_Report
 - Columns: Define by your own

Task 6: (15 min)

- Run your data pipeline to load cleansed and normalized data into final Data Marts;
- Export each individual data mart in csv file and put all files into archive;
- Count number of rows and distinct primary keys in each data Data Mart;
For the table with "Row Id" column also add distinct count of that field;

Deliverable 6:

- File 1 Name: Task_6_1_Data_Marts.zip
File Type 1: zip archive
- File 2 Name: Task_6_2_Data_Marts_Rows.csv
File 2 Type: csv
File 2 Structure: Data Mart System Name – Count Rows – Count Distinct Primary Key
(!For Table with Row Id field also calculate count distinct Row Id)

6. Summary

Task 7: (20 min)

- Push your key scripts to GitHub and share the link.

- Create a Read.me file explaining the steps you have considered during execution and why. You can add your suggestions and observations of data sets which you processed.

Deliverable 7:

- Link to GitHub repository with your scripts
- File Name: Task_8_GitHub_Link.txt
- File Type: txt or mhtml

List if deliverables and impact on evaluation:

Task #	Task Name	Deliverable	Impact on Evaluation
Task_1	KPI Glossary	Task_1_KPI_Glossary.csv	5%
Task_2	High Level Design	Task_2_HLD.pptx	15%
Task_3	Entity Relationship Diagram	Task_3_ERD.jpg	10%
Task_4	DWH DDLs	Task_4_DDL.txt	10%
Task_5	Inconsistencies Analysis and DQ report	Task_5_Inconcistencies_Analysis.xlsx	35%
Task_6	Data Marts	Task_6_1_Data_Marts.zip Task_6_2_Data_Marts_Rows.csv	15%
Task_7	Scripts in GitHub	Task_7_GitHub_Link.txt	10%